

Related Work

Robust Controller Synthesis Using PIE and IQC Duality

Robust control of distributed-parameter systems requires methods that account for both spatial dynamics and uncertainty. Established approaches include PDE backstepping, direct Lyapunov methods for uncertain parabolic systems, and semidefinite programming for PDE stability analysis [1–3]. The partial integral equation (PIE) representation provides an algebraic description of a class of PDEs through bounded partial integral (PI) operators [4, 5]. Its dual representation enables nominal state-feedback synthesis using operator inequalities [6]. Extending this construction to uncertain systems requires uncertainty descriptions compatible with both the PIE representation and controller synthesis.

Integral quadratic constraints (IQCs) provide a framework for describing uncertainty and nonlinear input–output behavior [7]. Their dissipation interpretation connects these descriptions to storage inequalities [8, 9]. PI multipliers have been used for robust ODE–PDE analysis [10], and hard IQCs with infinite-dimensional channels admit sufficient PIE-based KYP tests [11]. More recently, an IQC-based PIE framework has addressed structured robustness, performance, and observer-gain synthesis [12]. These results motivate a corresponding treatment of robust feedback controller synthesis.

IQC duality also has established finite-dimensional synthesis applications. Dual constraints enable robust disturbance-feedforward design [13], while dual filters and storage transformations support convex estimator and feedforward synthesis for gridded LPV systems [14]. Convexity depends on the plant and controller structure [15]. For more general dynamic multipliers, synthesis methods commonly alternate between controller design and multiplier analysis [16, 17]. Dynamic-IQC feedback constructions also exist for uncertain systems with state delay [18]. These distinctions are relevant when comparing a convex synthesis condition with a joint controller–multiplier optimization.

The passage between frequency-domain multipliers and finite-horizon inequalities requires particular care. Factorizations preserving graph and inverse-graph conditions are studied through doubly-hard factorizations [19]; finite-horizon IQCs with terminal costs provide a complementary connection to dissipativity [20]. Additional storage associated with the multiplier can also reduce conservatism [21, 22]. These results distinguish admissible finite-horizon certificates from frequency-domain factorization identities alone.

This paper studies robust state-feedback synthesis for uncertain systems admitting a PIE representation through duality of dynamically filtered IQC conditions. The construction addresses the placement of the reconstruction operator in the synthesis inequality and the recovery of controllers from dual operator variables. The synthesis problem is considered for fixed admissible filter data, with explicit causal inverse and realization assumptions. Static factors lead to state feedback, while dynamic factors introduce controller states driven by specified available signals. The uncertainty classes and measurement assumptions are stated separately for these two realizations.

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